MIL-F-51525A(EA)
3 February 1988
SUPERSEDING
MIL-F-51525(EA)
21 August 1987

MILITARY SPECIFICATION

FILTER, GAS, 200 CFM

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope. This specification covers one type of cylindrical gas filter with airflow rating of 200 cubic feet per minute (cfm).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants.

MILITARY

MIL-C-13724 - Charcoal, Activated, Impregnated, ASC.
MIL-F-51527 - Filter Set, Gas and Particulate, 200 CFM.

: Beneficial comments (recommendations, additions, deletions) and any perti- :
: nent data which may be of use in improving this document should be addressed:
: to: Commander, U.S. Army Chemical Research, Development and Engineering :
.Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using:
: the self-addressed Standardization Document Improvement Proposal (DD Form :
: 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance Test Methods.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-19-6368 - Filter, Gas.

(*Copies of specifications,* standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

STANDARD TEST METHOD

ASTM D 2867 - Moisture in Activated Carbon.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-6368 and on the subsidiary drawings shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings,

3.1.2 Components All components of the gas filter shall conform to the specifications and drawings listed on Drawing 5–19–6368 and subsidiary drawings.

3.2 Manufacture and assembly. The gas filter shall be manufactured and assembled in accordance with Drawing 5-19-6368.

3.3 Performance.

3.3.1 Airflow resistance. The airflow resistance of rhe gas filter shall not exceed 4.5 inches water gauge (iwg) at an airflow rate of 200 ± 10 cfm when tested in accordance with 4.4.4.1.

3.3.2 Leakage. The gas filter shall not leak when dichlorodifluoromethane refrigerant (R-12) conforming to type 12 of BB-F-1421 is introduced at the influent side of the filter as specified in 4.4.4.1 at the following test parameters:

Airflow rate	40 + 2 cfm
R-12 influent concentration	1000 Vppm
Airstream relative humidity (RH)	Less than *or* equal to 30 percent
Airstream temperature	75° ± 5°F

A leak is defined as the detection of more than 1 volume parts per million (Vppm) at the effluent side of the filter within 2 minutes after the introduction of R-12.

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3.3.3 Airflow resistance and dimethylmethylphosphonate (DMMP) gas-life 1/ after rough handling. The unpackaged gas filter (5-19-6368) shall be rough handled with the cylindrical axis of the filter parallel to the direction of the vibration input and the identification label located on the upper half of the filter. The test shall be run for 15 minutes at 3/4 inch amplitude and a frequency of 200 cycles per minute in accordance with 4.4.4.2. After rough handling, the gas filter shall show no evidence of cracks, dents, or carbon granule leakage; tile airflow resistance shall not exceed 4.5 iwg; and the DMMP gas-life shall be no less than SO minutes when tested in accordance with 4.4.4.3 at the following test parameters:

	DMMP
Percent purity (minimum)	99
Airflow rate	200 ± 10 cfm
lnfluent concentration 2/	5 ± 0.25 mg/L
Airstream temperature	110° + 5°F
Filter preheat	110″ ± 5°F
Break concentration 3/	0.04 ug/L

3.3.4 Cyanogen chloride (CK) gas-life. After completion of all manufacturing steps, the gas filters carbon shall comply with the CK gas sorption requirement of MIL-C-13724 for grade IV unaged carbon when tested in accordance with 4.4.4.5.

3.3.5 Moisture content. The moisture content of the carbon in the filter, at the time of packaging, shall not exceed 2 percent by weight when tested in accordance with 4.4.4.4.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of gas filters shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The gas filter shall be free of abraded gaskets and foreign matter (dirt, oil, or viscous materials).

l/The DMMP gas-life is defined as the time from the introduction of DMMP at the influent side of the filter to the detection of the specified break concentration at the effluent side of the filter.
2/Influent concentration is in milligrams per liter (mg/L).
3/Break concentration is in micrograms per liter (ug/L).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government responsibility. Unless otherwise specified, the Government will be responsible for t-he performance of t-he tests specified in 4.4.4.2, 4.4.4.3 and 4.4.4.5. Samples will be forwarded to the laboratory designated by the contracting officer (6.2).

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Prcproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 preproduction inspection.

4.3.1 Lotting. A preproduction lot of 20 gas filters (see 6.2(d)) shall be manufactured and packaged in accordance with MIL-F-51527 using the same methods, materials, equipment, and processes as will be used during regular production (see 6.2).

4.3.2 Sampling.

4.3.2.1 For examination. Prior to packaging in accordance with MIL-F-51527 each gas filter in the preproduction lot shall be examined in accordance with 4.3*3.1.

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4.3.2.2 For nondestructive tests. Sampling for nondestructive tests shall be performed prior to packaging in accordance with MIL-F-51527.

(a) Airflow resistance and R-12 leakage. Each gas filter in the preproduction lot shall be tested for airflow resistance and R-12 leakage.

(b) Moisture content. Four sample gas filters in the preproduction lot shall be tested for moisture content.

(c) CK gas-life. A sample of carbon (500 milliliter (mL) minimum) shall be taken during the filling of four filters. Each filter shall be randomly chosen from a different quarter of the preproduction lot. Each sample shall be placed in an open container or in a cloth bag, and shall accompany its filter through the assembly line (the sample need not be exposed to R-12). Upon filter packaging (see section 5), each sample shall be placed in an airtight container and submitted for test.

4.3.2.3 For destructive tests. Sampling for destructive tests of the gas filter shall be performed after packaging into filter set in accordance with MIL-F-51527.

4.3.2.3.1 Airflow resistance and DMMP gas-life after rough handling. Eight sample gas filters shall be taken randomly from the preproduction lot.

4.3.3 Inspection procedures.

4.3.3.1 For examination. Prior to packaging in accordance with MIL-F-51527 each gas filter shall be examined for the defects listed in the classification of defects (4.4.3.4). The AQLs listed in the classification of defects do not apply; if any filter is found to be defective, the preproduction lot shall be rejected.

4.3.3.2 For nondestructive tests. Nondestructive tests on gas filters shall be performed prior to packaging in accordance with MIL-F-51527.

(a) Airflow resistance and R-12 leakage. Each filter shall be tested for airflow resistance and R-12 leakage in accordance with 4.4.4.1. Any filter failing to meet the requirements of 3.3.1 *or* 3.3.2 shall be rejected.

(b) Moisture content. Sample gas filters shall be tested for moisture content in accordance with 4.4.4.4(a) and 4.4.4.4(b). This test shall be run immediately before packaging the gas filter. The results of these tests shall meet the criteria of MIL-STD-414 at an AQL level of 1.0 percent defective for the preproduction lot to be acceptable.

(c) CK gas-life. CK gas-life shall be performed in accordance with 4.4.4.5.

4.3.3.3 For destructive tests. Destructive tests on the gas filter shall be performed after the filter set as been unpackaged and the gas filter removed from the filter set.

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4.3.3.3.1 Airflow resistance and DMMP gas-life after rough handling. Sample gas filters shall be rough handled in accordance with 4.4.4.2 and then tested for airflow resistance and DMMP gas-life in accordance with 4.4.4.3.

4.3.4 Acceptance/rejection criteria. The preproduction lot shall comply with all requirements of this specification when examined in accordance with 4.3 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the gas filter assemblies of one type produced by one manufacturer, from the same materials, at the same location, under essentially the same manufacturing conditions, and at essentially the same time. However, no more than one lot of carbon sorbent shall be represented in any one lot of finished filters. Each lot of filters shall be divided into sublets of one day's production each.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling for all examinations listed in the classification of defects shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For nondestructive tests.

(a) Airflow resistance and R-12 leakage. Each filter shall be tested for airflow resistance and R-12 leakage.

(b) Moisture content. Sampling for moisture content, 4.4.4.4(a), shall be the first filter of the day, every ninth filter, and the last filter of the day if more than four filters have been fabricated since the last sample. If one barrel of carbon is used at a time for filling filters, at least 2 samples shall be taken from each barrel, one at the beginning and one at the end.

(c) CK gas-life. A sample of carbon (500 mL minimum) shall be taken during the filling of the first filter, every fiftieth filter, and the last filter of the day. The sample shall be placed in an open container or in a cloth bag and shall accompany its filter through the assembly line (the sample need not be exposed to R-12). Upon filter packaging (see section 5), the sample shall be placed in an airtight container. Upon completion of the'lot, a sample shall be randomly chosen from each quarter of the lot. These four samples shall be submitted for test.

4.4.2.3 For destructive tests. Sampling for destructive tests shall be performed after packaging the gas filter into the filter canister.

4.4.2.3.1 Airflow resistance and DMMP gas-life after rough handling. Sampling shall be conducted in accordance with MIL-STD-414, level II, AQL 1.0 percent.

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4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample gas filters shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For nondestructive tests.

(a) Airflow resistance and R-12 leakage. Each filter shall be tested for airflow resistance and R-12 leakage in accordance with 4.4.4.1. Any filter failing to meet the requirements of 3.3.1 and 3.3.2 shall be rejected.

(b) Moisture content. Immediately before packaging the moisture content of each filter shall be determined in accordance with 4.4.4.4(a) and 4.4.4.4(b). Any filter that does not meet the moisture content requirement shall be removed from the lot and dried before packaging (see 4.4.4.4(b)).

(c) CK gas-life. The samples shall be tested for CK gas-life in accordance with 4.4.4.5. The production lot of filters shall be rejected if any sample fails to meet the requirement.

4.4.3.3 For destructive tests. Destructive tests on the gas filters shall be performed after the filter set has been unpackaged and the gas filter removed from the filter set.

4.4.3.3.1 Airflow resistance and DMMP gas-life after rough handling. Sample gas filters shall be tested for airflow resistance and DMMP gas-life in accordance with 4.4.4.3. The production lot of filters shall be rejected if any filter fails to meet the requirements of 3.3.3.

4.4.3.4 Classification of defects.

(a) Filter, gas (Drawing 5-19-6368).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing, incorrect, or incorrectly assembled
102	Overall dimensions incorrect
103	Information plate missing or illegible
104	Gasket damaged
105	Bulges or dents more than l/4-inch
Minor:	AQL 2.5 percent defective
201	Workmanship (3.5)

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4.4.4 Tests. Test on the gas filter shall be conducted as follows:

4.4.4.1 Airflow resistance and R-12 leakage. Place the gas filter in a test fixture. Regulate the air source of the influent side of the filter to 200 cfm and record the airflow resistance of the filter. Note: The test fixture resistance must be subtracted so that the recorded resistance is that of the filter only. Readjust the airflow to 40 cfm and the required temperature and RH. Introduce the specified concentration of R-12 at the influent side of the filter. After two minutes, record the concentration of R-12 at the effluent side of the filter (see 6.4). The R-12 concentration shall not exceed the specified maximum.

4.4.4.2 Rough handling. The unpackaged filter (5-19-6368) shall be rough handled with the cylindrical axis of the filter parallel to the direction of vibration input. The filter shall be oriented in the test fixture such that the identification label is in the up position, Vibrate filter in accordance with test method T 105.10 of MIL-STD-282 and the rough handling parameters specified in the applicable requirement. Inspect filter for cracks, dents, or carbon leakage.

4.4.4.3 Airflow resistance and DMMP value after rough handling. After the filter has been rough handled in the vertical position for 15 minutes as specified in 4.4.4.2, the sample filter shall be tested for DMMP value (see 6.5 and 6.6) in an apparatus which establishes the following test comditions:

(a) Preheat sample filters to the specified temperature in a forced draft oven for 30 minutes.

(b) Insert an expendable filter, not a sample filter for test, in the test apparatus and start the blower.

(c) Adjust the controls until the rated airflow and temperature are reached.

(d) Generate DMMP until the specified concentration is reached. Note the generator and systems settings.

(e) Shut down the tester, remove the expendable filter, and insert the preheated sample filter to be tested.

(f) Start the tester without DMMP and quickly bring the tester to the proper flow and temperature.

(g) Start the flow of DMMP and quickly bring the concentration to the specified level.

(h) Start the timer when the influent concentration detector first indicates DMMP in the system. Record the airflow resistance (iwg) of the filter. Note: The test fixture resistance must be subtracted so that the recorded resistance *is* that of the filter only.

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(i) Stop the test and record the time when the effluent concentration detector indicates that the break concentration has been reached.

4.4.4.4 Moisture content.

(a) Determination of filter dry weight. A 10 gram sample shall be taken during the leveling operation of the sample gas filters of each sublet, and the weight of carbon in each filter shall be measured to the nearest 1/2 ounce immediately after leveling the carbon. The moisture content of the carbon samples shall be determined according to ASTM D 2867 and the results averaged. This percent weight loss value is defined as the moisture content of the represented sublet of gas filters. The weight of carbon value of each filter in the sublet shall be reduced by this percentage to obtain the dry weight of the carbon and this weight shall be recorded on the information plate on each filter. The filter dry weight shall also be entered at this time.

(b) Determination of moisture content. Immediately before packaging, the filter shall be weighed to the nearest 1/2 ounce and the weight gain, if any, shall be noted. This weight gain shall be less than tile acceptable. If the filter does not meet this requirement, it may be dried at a temperature not to exceed 150°F and retested.

4.4.4.5 CK gas-life. A 250 + 5 mL sample of carbon shall be removed from each of the four samples. These samples shall be tested for CK gas-life in accordance with MIL-C-13724. Each of the samples must meet the minimum requirements.

5. PACKAGING

5.1 Level A. Preservation, packing, unit packing and marking shall be in accordance with MIL-F-51527 (see 6.3).

6. NOTES

6.1 Intended use. This specification covers a 200 cfm gas filter intended for use with Modular Collective Protection Equipment (MCPE).

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packing required.
(c) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether to not to proceed with production.

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(d) Deliverable preproduction filters. The filters not subjected to destructive testing may be considered deliverable items provided that the pre-production lot is accepted. Appropriate provisions should be made to deliver these filters to the designated activity.

6.3 Level A packaging. The gas filter covered by this specification is obtained packaged with a particulate filter in accordance with MIL-F-51527.

6.4 R-12 leakage detector. The halogen leak detector model HLD-2 by Inficon Leybold-Haraeus Inc. has been found suitable for this purpose.

6.5 DMMP effluent concentration detector. The C.S.I. Phosphorus Analyzer, Model No. PA260, has been found suitable for this purpose.

6.6 DMMP influent concentration detector. The Beckman Model 400 Hydrocarbon Analyzer has been found suitable for this purpose.

6.7 Subject term (key word listing).

Airflow resistance
Carbon, activated
Charcoal, activated
Filter, gas
Filter gas life

6.8 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-51525A(EA), Amdt 7	2. DOCUMENT DATE (YYMMDD) 920710

3. DOCUMENT TITLE
FILTER, GAS, 200 CFM

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If Applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. Army Chemical Research, Development and Engineering Center	b. TELEPHONE (Include Area Code) (1) Commercial: (301) 671-3259 (2) AUTOVON: DSN 584-3259
c. ADDRESS (Include Zip Code) Commander U.S. Army CRDEC ATTN: SMCCR-PET-S Aberdeen Proving Ground, MD 21010-5423	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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